

IGNISIA

- Theme: Smart Cities
- Project Title: AI-Powered Smart Society Security Monitoring System
- Team: Cypher (Team ID – **1233**)

Problem Statement: AI-powered surveillance system that automatically monitors CCTV feeds, detects threats in real time, identifies residents using face recognition, and sends instant alerts when suspicious activity is detected.



Introduction

Problem Statement

- Residential societies face security **challenges** such as **unauthorized access, theft**, and difficulty in continuously monitoring **CCTV** feeds.
- Traditional security systems rely heavily on **manual monitoring**, which is slow, costly, and prone to human error.

Need & Background

- There is a growing need for **automated** and **intelligent** surveillance systems.
- AI technologies like **computer vision and face recognition** can analyze CCTV footage in **real time** to detect people, identify residents, and flag suspicious activities.

Existing solutions & their limitations

- Traditional CCTV requires constant human monitoring
- Security guards cannot monitor multiple cameras efficiently
- Most systems lack real-time alerts and AI-based threat detection
- Existing setups cannot differentiate residents from unknown individuals



AUTOMATIC CCTV
MONITORING



REAL-TIME THREAT
DETECTION



RESIDENT FACE
RECOGNITION



INSTANT ALERTS &
NOTIFICATIONS

Introduction

Our Solution

Tech Approach

Feasibility

Impact

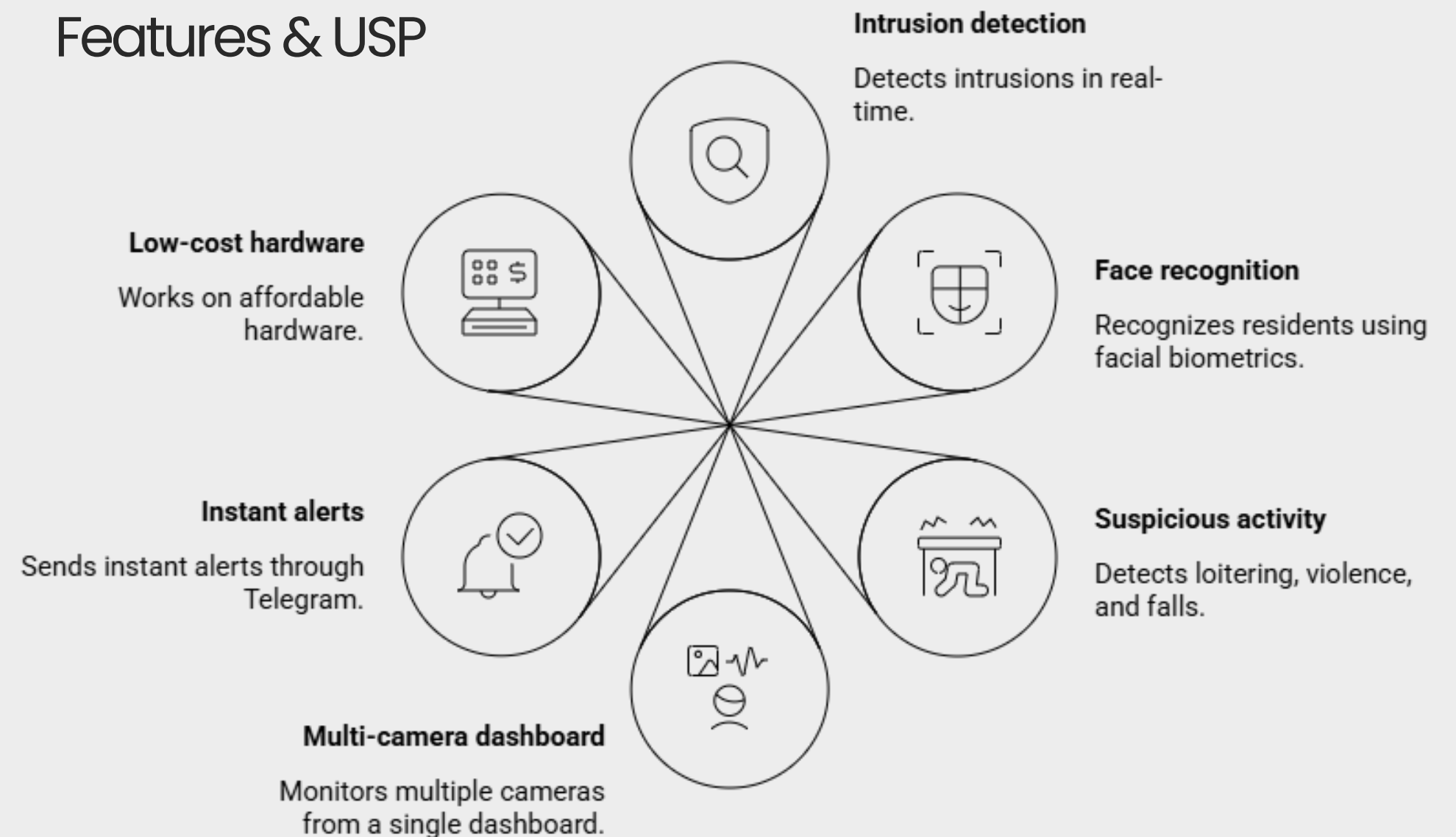
Research

Our Solution

* Solution Description

- AI-powered surveillance system that transforms traditional CCTV into an **intelligent security platform**.
- Reduces the need for continuous manual monitoring.
- Uses **computer vision** and machine learning to analyze multiple camera feeds in real time.
- Detects people, recognizes residents, and **identifies suspicious activities** through an AI pipeline.
- Sends **instant Telegram alerts** with snapshot, camera name, alert type, and timestamp.
- Runs in under **one second per frame** on standard CPU hardware without cloud dependency.

Features & USP



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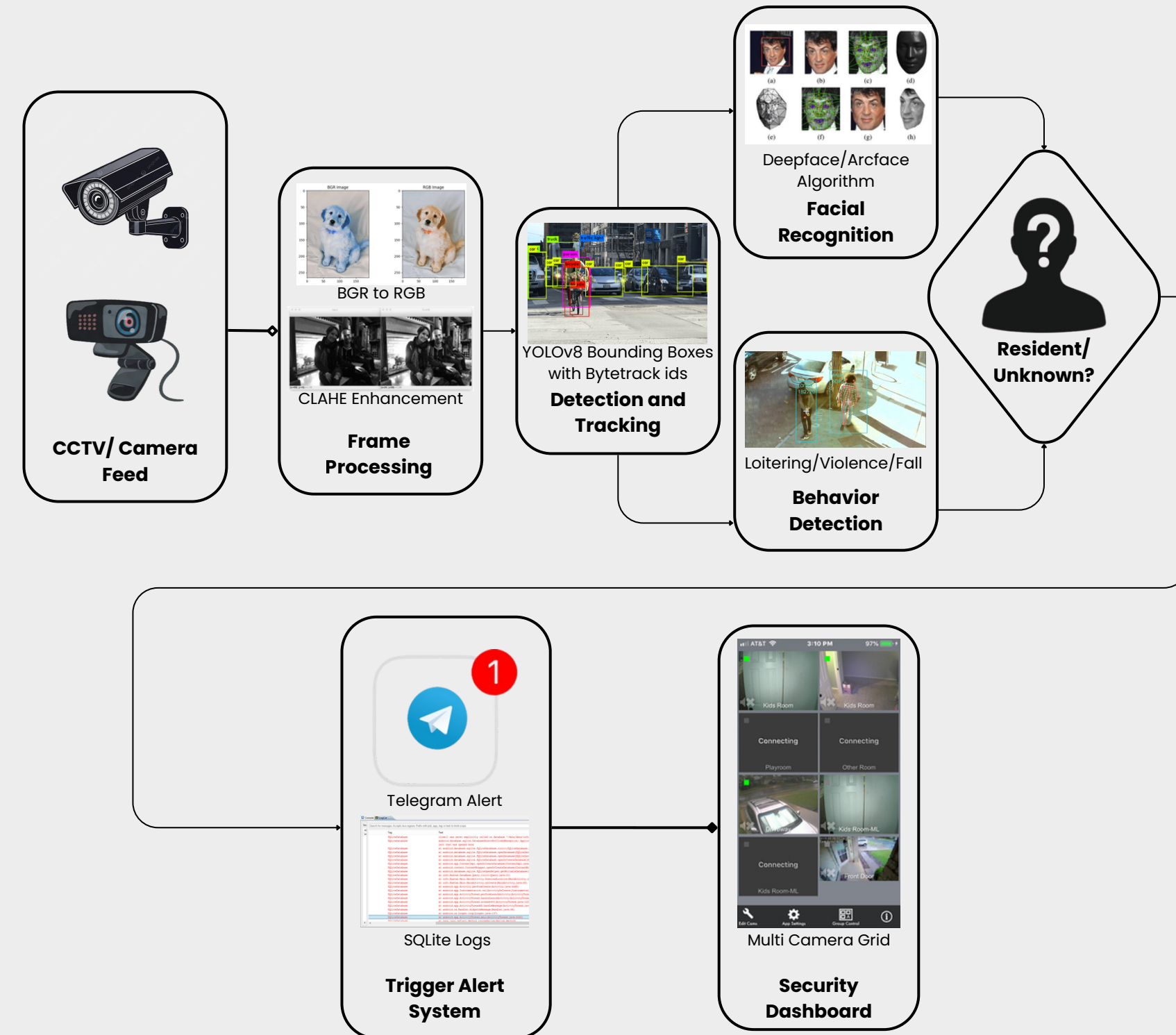
Technical Approach

Methodology

- 01 Camera Ingest**
Live CCTV / IP camera / webcam feeds via OpenCV
- 02 Detect & Track**
YOLOv8n detects persons; ByteTrack assigns track IDs
- 03 Parallel AI**
Face recognition + behavior analysis run concurrently
- 04 Evaluate**
Alert conditions checked against cooldown registry
- 05 Alert & Log**
Telegram snapshot + SQLite event log

Implementation

-  **Person Detection & Tracking**
YOLOv8n + ByteTrack
-  **Face Recognition**
DeepFace + ArcFace (512-dim)
-  **Behavior Detection**
RWF-2000 CNN+LSTM + MediaPipe Pose
-  **Alert System**
Telegram Bot + SQLite3 logging
-  **Multi-Camera Dashboard**
OpenCV threaded grid + RTSP support



SystemFlow



Introduction

Our Solution

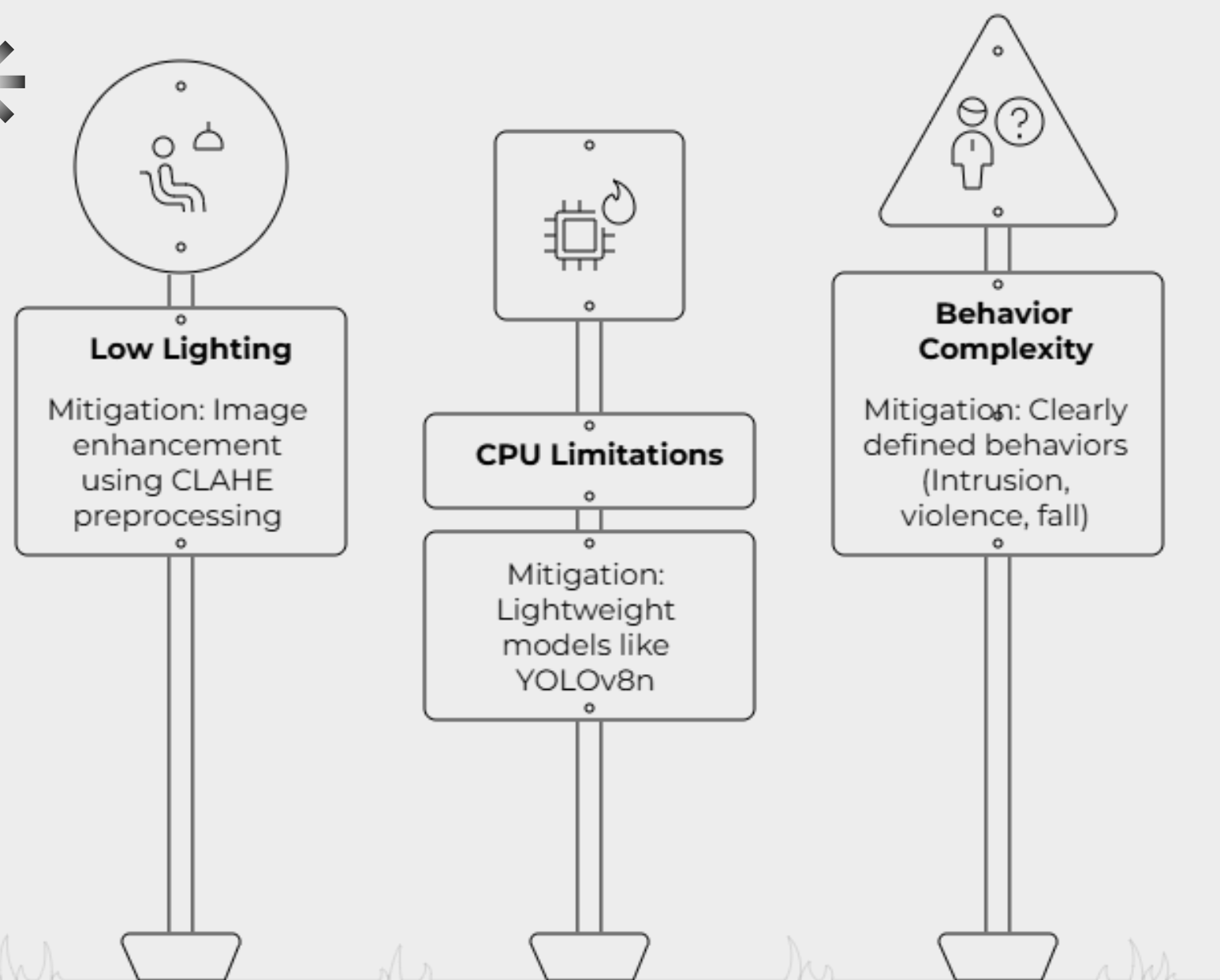
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Feasibility & Viability



Viability & Scalability

Deployment Cost

Hardware (1 IP camera)
Compute (existing laptop/PC)
Software (all open-source)
vs. Commercial AI CCTV system

₹1,500–2,500
₹0 additional
₹0
₹5–15 Lakh

Scalability

- System can easily scale from a single camera to hundreds of cameras.
- Cloud or GPU servers can process large-scale surveillance networks.
- Modular architecture allows adding new AI modules like weapon detection, fire detection, or crowd analysis.
- Suitable for residential societies, campuses, offices, malls, and smart cities.



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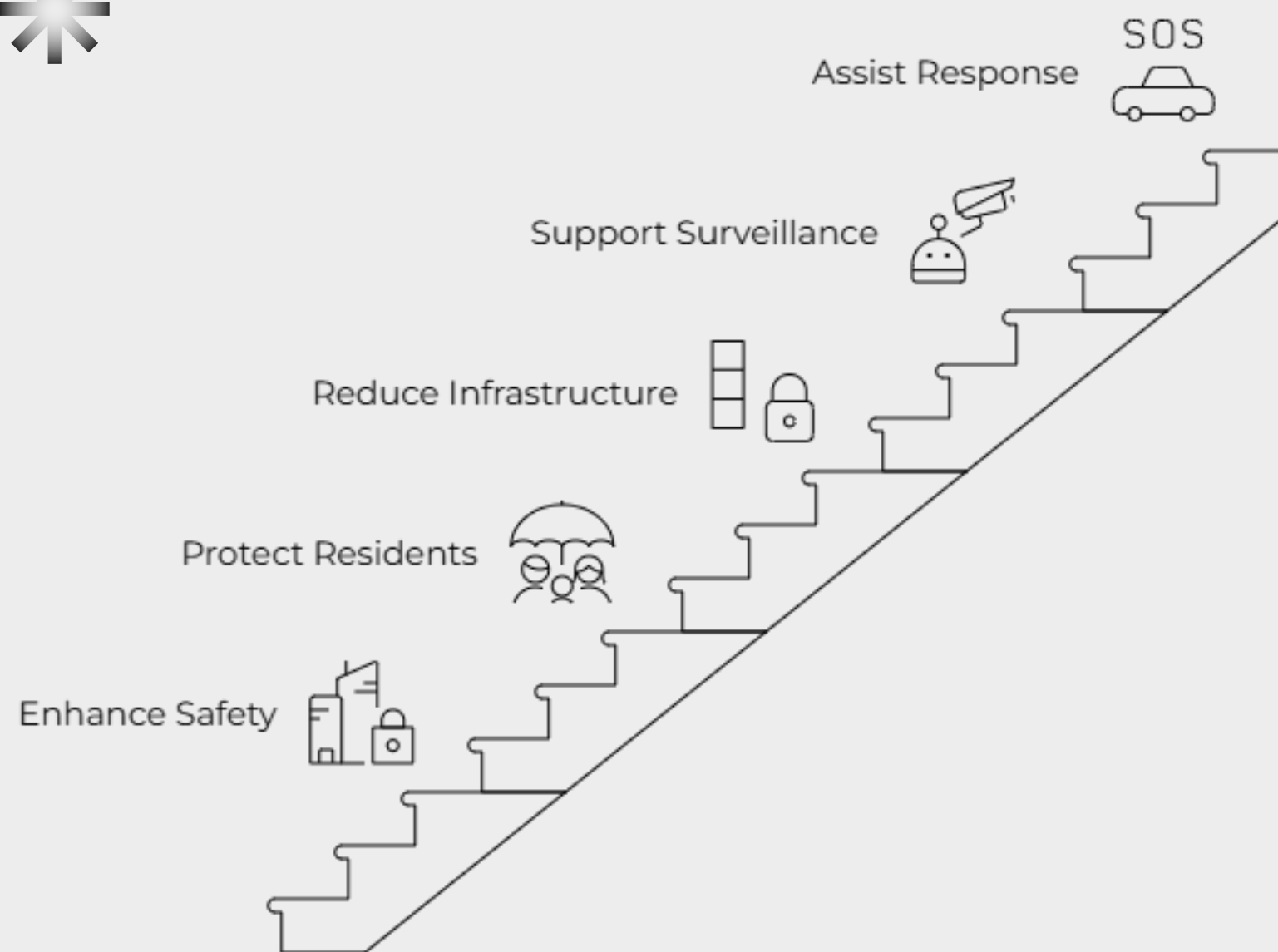
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Benefits

vs. Manual Guards

✓ Ours: 24/7 automated, no fatigue, no salary

✗ Current: ₹15K+/mo, limited coverage

vs. Basic CCTV

✓ Ours: Real-time detection + alerts + analytics

✗ Current: Passive recording only

vs. Commercial AI

✓ Ours: Open-source, ₹2K hardware, fully customizable

✗ Current: ₹5-15L, vendor locked

<10s

Alert Response Time

~₹2K

Total Hardware Cost



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Research & References

- **YOLOv8** – Ultralytics Object Detection
- **DeepFace** Face Recognition Framework
- **MediaPipe** Pose Estimation
- **RWF-2000** Violence Detection Dataset
<https://github.com/mchengny/RWF2000-Video-Database-for-Violence-Detection>
- **OpenCV** Computer Vision Library

- [1] Ultralytics, "YOLOv8 Docs," 2023. [Online]. Available: <https://docs.ultralytics.com/>. [Accessed: Mar. 17, 2026].
- [2] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in Proc. Eur. Conf. Comput. Vis. (ECCV), 2022. [Online]. Available: <https://arxiv.org/abs/2110.06864>.
- [3] S. I. Serengil and A. Ozpinar, "DeepFace: A lightweight face recognition and facial attribute analysis library for Python," in Proc. Innov. Comput. Commun. (ASIC), 2020. [Online]. Available: <https://github.com/serengil/deepface>.
- [4] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 4690–4699.
- [5] V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann, "BlazePose: On-device real-time body pose tracking," arXiv preprint arXiv:2006.10204, 2020. [Online]. Available: <https://ai.google.dev/edge/mediapipe>.
- [6] M. Cheng, K. Cai, and M. Li, "RWF-2000: An open large scale video database for violence detection," in Proc. 25th Int. Conf. Pattern Recognit. (ICPR), 2020, pp. 4183–4190. [Online]. Available: <https://arxiv.org/abs/2002.04367>.



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